VISION-LANGUAGE-ACTION (VLA) ARCHITECTURE & PROPOSAL PIPELINE

Multimodal Ingestion · Visual Patch Tokenization · Autoregressive Transformer Backbone · Action Tokenization Paradigms

1. MULTIMODAL INGESTION

Overhead RGB Camera
224 x 224 x 3 @ 30 Hz
ViT Patch: 14x14 -> 256 tokens
Linear Projection z_vis

Wrist RGB Camera
224 x 224 x 3 (Gripper View)
Fine-contact visual alignment
Linear Projection z_wrist

Task Instruction (Text)
"pick red cup, place on tray"
BPE Tokenizer / SentencePiece
Text Tokens: [t_1, t_2, ..., t_L]

Proprioceptive State
Joint Angles q, Velocities q_dot
Gripper state g in [0, 1]
State Token: z_state

2. VLA TRANSFORMER BACKBONE (7B–55B)

Multimodal Fusion & Self-Attention
Concatenated Tokens: [z_vis | z_wrist | z_text | z_state]
32–80 Transformer Layers · Multi-Head Attention
Web-scale pretraining + Robot Trajectory co-fine-tuning
Exemplars: OpenVLA (7B), RT-2 (55B), PaLM-E (562B)
Memory Bound: Streams ~7 GB weights per action step
Inference Latency: 150–350 ms on Edge NPU (3–5 Hz)

Edge Silicon Resource Footprint
Precision: INT4 / FP8 Quantized Weights
Memory: 5.6 GB DRAM (Weights) + 1.2 GB KV Cache
Bus Bandwidth: 204.8 GB/s LPDDR5 Memory Wall
Thermal Draw: 35W–50W sustained on SoC
Hardware: NVIDIA Jetson AGX Orin / Apple Silicon

UNPRIVILEGED CANDIDATE PROPOSALS
Outputs are stochastic samples a_hat ~ pi_theta(o_t)
ZERO direct actuator register write authority

3. ACTION TOKENIZATION & EMISSION

Paradigm A: Discrete Action Binning (RT-2 / OpenVLA)
Quantize 7-DoF action space into K=256 uniform bins
Delta coords: [dx, dy, dz, droll, dpitch, dyaw, grip]
Emits 7 text vocabulary tokens sequentially
Single-step delta proposal: a_t in R^7 @ 3–5 Hz
Artifact: Discrete Waypoint / Single-Step Goal

Paradigm B: Diffusion Action Chunking (Octo / ACT)
Transformer features condition lightweight diffusion head
Predicts continuous action chunk A_t:t+H (H = 16 steps)
Temporal ensembling eliminates single-step Markov error
Continuous trajectory horizon: 320 ms preview window
Artifact: C^2 Continuous Spline / Action Chunk

4. Downstream Nervous System Gating (MCU @ 1 kHz)
RPMSG / PCIe DMA across isolation boundary
Control Barrier Function (CBF-QP): h_dot(x,u) + gamma h(x) >= 0
Stopping distance invariant: d_stop(v) <= d_clearance
Permitted Motor Commits u_t* Latch at 1000 Hz
Fallback: Safe Stop 1 (SS1) / Zero-torque if lease expires

THE PHYSICAL AI DILEMMA: HIGH CAPACITY (7B–55B) INFERENCE RUNS SLOW (3–5 Hz); PHYSICAL DYNAMICS REQUIRE FAST (1 kHz) SUPERVISION

VLAs operate strictly as unauthenticated proposal engines across the slow cognitive tier, gated by bare-metal MCU barrier enforcers.